Iris: the burnout stability prediction was too high

Preferred 0.50-inch transition

At 20.77 s	CP (m)	Margin (cal)
Legacy OpenRocket	2.239	0.11
Full finite wing	2.644	3.91
Scalar RASAero fit	2.468	2.26
DATCOM blend	**2.573**	**3.24**
RASAero II	2.510	2.65

The discrepancy is in CP. CG agrees at 2.2273 m. The full correction places CP 134 mm too far aft relative to RASAero; the new blend cuts that to 63 mm.

Recommendation. Use the new option to compare designs. Keep the native RASAero result alongside it: the blend still overpredicts margin by 0.59 cal.

Different data answer different questions

Dataset	Question it answers
5 fitted rockets	What normal-force and first-moment adjustment minimizes CP error at 0°, Mach 1.50–3.00?
7 reserved + 3 first blind	Does a calibration transfer to other spans, sweeps, tapers, fin counts and body diameters?
5 bare + 7 variants	Which discrepancies survive without fins? What changes with section, scale or CP setting?
2 tunnel vehicles	Does tuning to another code damage agreement with measured CP or normal force?
3 final blind + Iris	How does the literature-based model perform on newly frozen geometries and in an actual two-stage simulation?

Measurement convention

CP is distance aft of the nose. Error is OpenRocket minus reference, in active body diameters. RASAero is an independent prediction; tunnel data are physical measurements.

Separate fin loading from forebody pitching moment

What changed in the research build

The original option integrates pressure on a finite trapezoid. The new branch also uses USAF DATCOM tables for fin slope and nose–cylinder potential loading. It blends normal force and first moment before dividing for CP.

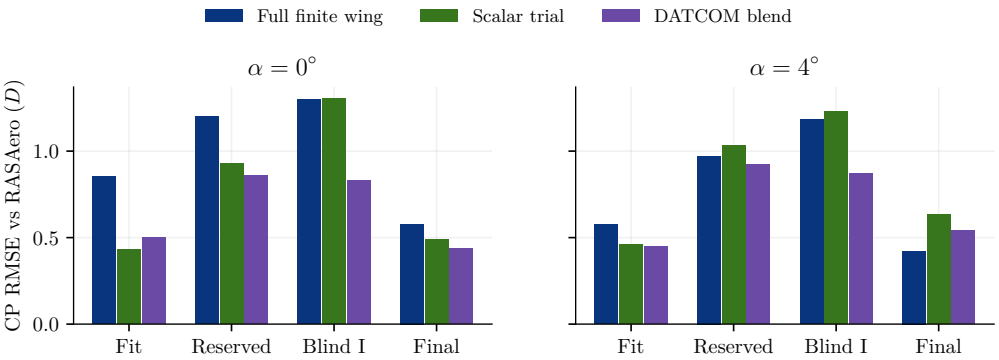
The chosen weights use 32% of the change to the chart fin model and 40% of the chart forebody correction. Thirty geometries exercise the implementation at 44,190 states.

Why one multiplier failed

At Mach 2.5 and 4°, the H baseline's added fin-load slope is 4.59/rad in the full option and 4.61/rad in RASAero. Reducing the fin load spoils that agreement.

The independently fitted 45.9% scalar moves Iris closer to RASAero but degrades both tunnel CP benchmarks. We rejected it as a general replacement.

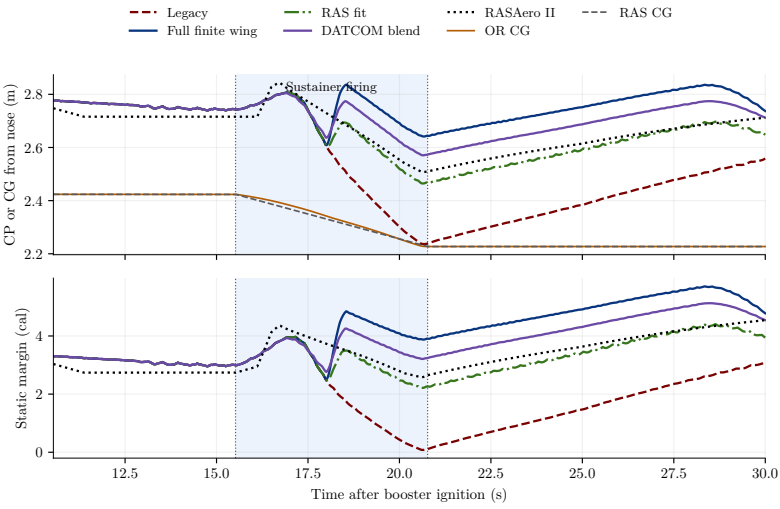
The new blend improves the final test at zero incidence



Final three, 0°: 0.58 to 0.44 *D* RMS. The strongest improvement is on the highly tapered rocket.

Counterexample: at 4° the final group goes from 0.42 to 0.54 *D* RMS. The near-rectangular rocket also worsens at 0°.

Iris: lower error through the burn, but the minimum is still early



Burn-wide margin RMS
Full: 0.95 cal
DATCOM blend: 0.56 cal

Minimum margin
Blend: 2.77 cal at 18.00 s
RASAero: 2.58 cal at 20.60 s

At the same Mach, angle and stage, the blend still places CP 0.68 *D* aft on average above Mach 2 in the sampled sustainer segment.

Use the blend for sensitivity studies; preserve both baselines

Physical regression checks

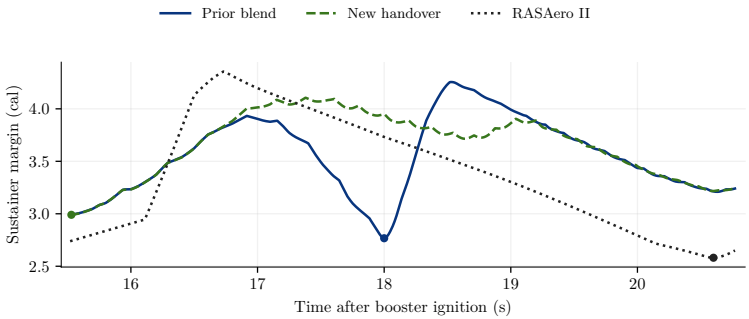
CP RMS in body diameters; five supersonic measurements per vehicle.

	ARCAS	Yuska B
Legacy	2.665	1.023
Full	.318	.126
Scalar	1.047	.511
DATCOM blend	.312	.136

Scope. The tunnel benchmarks constrain the blend. They are not independent evidence of its physical accuracy. ARCAS pitching-moment error increases despite nearly unchanged CP RMS.

Independent check. NACA TN 3795 provides separate body, wing and assembly measurements. The following tests compare their lift and moment slopes.

A smoother fin handover improves RMS and moves the minimum earlier



Burn-wide margin RMS falls from 0.56 to 0.44 cal.

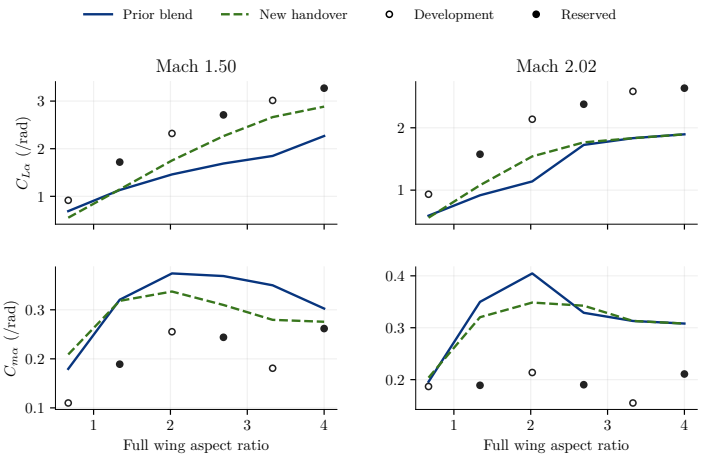
The new minimum occurs at 15.53 s. RASAero reaches its minimum at 20.60 s.

Burnout remains about 3.24 cal, compared with 2.65 in RASAero.

Iris component probes near burnout at Mach 2.80

Low-incidence probe slopes assign $+0.43\,D$ to the aft airframe, $+0.11\,D$ to the installed fins, and $+0.04\,D$ to the bare forebody. The accelerating and decelerating passes agree to $0.003\,D$ per term.

Measured wing–body interference exposes a remaining model error



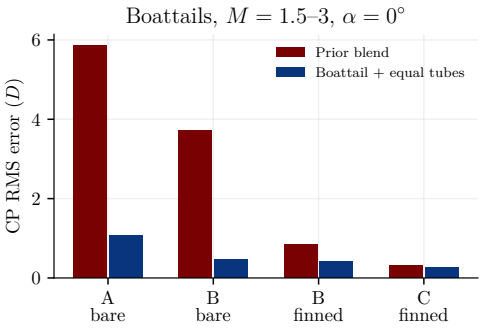
NACA TN 3795

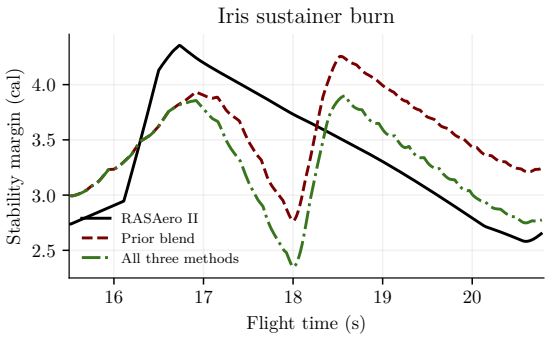
Three reserved wings at two Mach numbers. Assembled lift-slope RMS falls from 0.80 to 0.55/rad.

The isolated interaction worsens. Measured mean lift interference is -0.26/rad versus -0.79/rad in the new model.

Keep the prior research blend. The handover experiment identifies the crossover behavior and the aft-airframe load discrepancy.

Contracting boattails improve across three RASAero rocket families





Boattails. The reserved four-fin rocket improved at every tested incidence. At 2°, RMS error in the moment added by its taper fell from 0.076 to 0.008. ARCAS tunnel CP error rose from 0.31 to 0.36D.

Iris. The stepped-center model brought burnout from 3.24 to 2.77 cal. RASAero gave 2.65. The modeled minimum fell to 2.35 cal at 18.00 s. RASAero gave 2.58 cal at 20.60 s.

Which components merit a change in MIT 6.6?

Join identical tubes

Drawing the same 48-inch cylinder as four tubes moved CP by up to $0.70D$ in the prior blend. Using its continuous length eliminated the difference at all 453 tested conditions.

Evaluate contracting boattails

Use the DATCOM normal-force increment and its own local load station. Agreement with native RASAero improved on both bare bodies and both finned boattails.

Keep the other changes experimental

The stepped-center model increased finless Iris CP error from 2.86 to $4.16D$ at zero incidence. The tested flare correction worsened the measured flare-minus-cylinder moment. The crossflow change gave mixed finite-incidence results.

Verification. Thirteen fresh native static models, three archived Iris geometries, two native field diagnostics, three tunnel datasets and a full two-stage Iris rerun in the Windows VM.